

1 Annex: formulas and flowcharts

1.1 Survival probability per life stage

$$\text{logit}(p_{\text{surv,stage}}) = \beta_{0,\text{stage}} + \beta_1 s + e_{\text{stage}} \quad (1)$$

$$e_S = -\frac{100 - \text{RH}}{100} + f(I_{\text{rel}}) \quad (2)$$

$$e_J = 0.5 f(I_{\text{rel}}) \quad (3)$$

$$e_A = -\max(0, (T - 23) 0.05) \quad (4)$$

$$f(I_{\text{rel}}) = \gamma \min(I_{\text{rel}}, 1) - \delta \max(0, I_{\text{rel}} - 1.5) \quad (5)$$

where:

- s = pseudobulb size (cm)
- I_{rel} = relative radiation
- RH = relative humidity (%)
- T = temperature (°C)

1.2 Transition probability

$$\text{logit}(p_{\text{trans}}) = \psi_{0,\text{stage}} + \beta_{\text{precip}} P + \beta_{\text{rh}} \text{RH} \quad (6)$$

- p_{trans} = probability of transitioning between stages
- $\psi_{0,\text{stage}}$ = stage-specific intercept
- β_{precip} = precipitation effect coefficient
- β_{rh} = relative humidity effect coefficient
- P = precipitation
- RH = relative humidity

1.3 Flowering probability

$$p_F = \frac{1}{1 + \exp[-(\alpha_0 + \alpha_1 s)]} \quad (7)$$

- p_F = probability of flowering
- s = pseudobulb size (cm)
- α_0 = intercept controlling the flowering threshold
- α_1 = size effect coefficient

1.4 Fruit number

$$E[N_{\text{fruits}} \mid s] = \exp(\rho_0 + \rho_1 s) \quad (8)$$

- $E[N_{\text{fruits}} \mid s]$ = expected fruit number for a plant of size s
- ρ_0 = intercept of the fruit-count model
- ρ_1 = size effect coefficient for fruit production

1.5 Dispersal kernel

Mean dispersal distance decreases with height within the canopy, following ?], adapted from the canopy wind-attenuation term of ?]:

$$\bar{d}(h) = \min\left(\max\left(1, \text{round}\left(\frac{\bar{u} e^{a(h-h_c)/h_c}}{\lambda U_t}\right)\right), d_{\max}\right) \quad (9)$$

Individual seed displacement is drawn stochastically:

$$r \sim \text{Exponential}(\text{rate} = 1/\bar{d}), \quad r \leq d_{\max} \quad (10)$$

$$\theta \sim \theta_{\text{wind}} + \text{Uniform}\left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \quad (11)$$

$$x' = x + \text{round}(r \sin \theta) \quad (12)$$

$$y' = y + \text{round}(r \cos \theta) \quad (13)$$

$$z' = z + \text{DiscreteUniform}\{-z_{\max}, \dots, z_{\max}\} \quad (14)$$

- $\bar{d}(h)$ = mean dispersal distance at height h (grid cells)
- $\bar{u}$ = mean wind speed above canopy
- a = canopy attenuation coefficient, $a = 4 \phi_{\text{dif}}$ (rescaled to ? 's reported 0–4 range; ϕ_{dif} is the diffuse radiation fraction, itself used as a proxy for canopy density in the absence of a TCMF-specific wind coefficient)
- h_c = canopy height
- λ, U_t = terminal velocity scaling parameters
- $d_{\max}$ = maximum dispersal distance (grid cells)
- r = dispersal distance drawn per seed
- θ_{wind} = mean wind direction (radians)
- $z_{\max}$ = maximum vertical dispersal offset

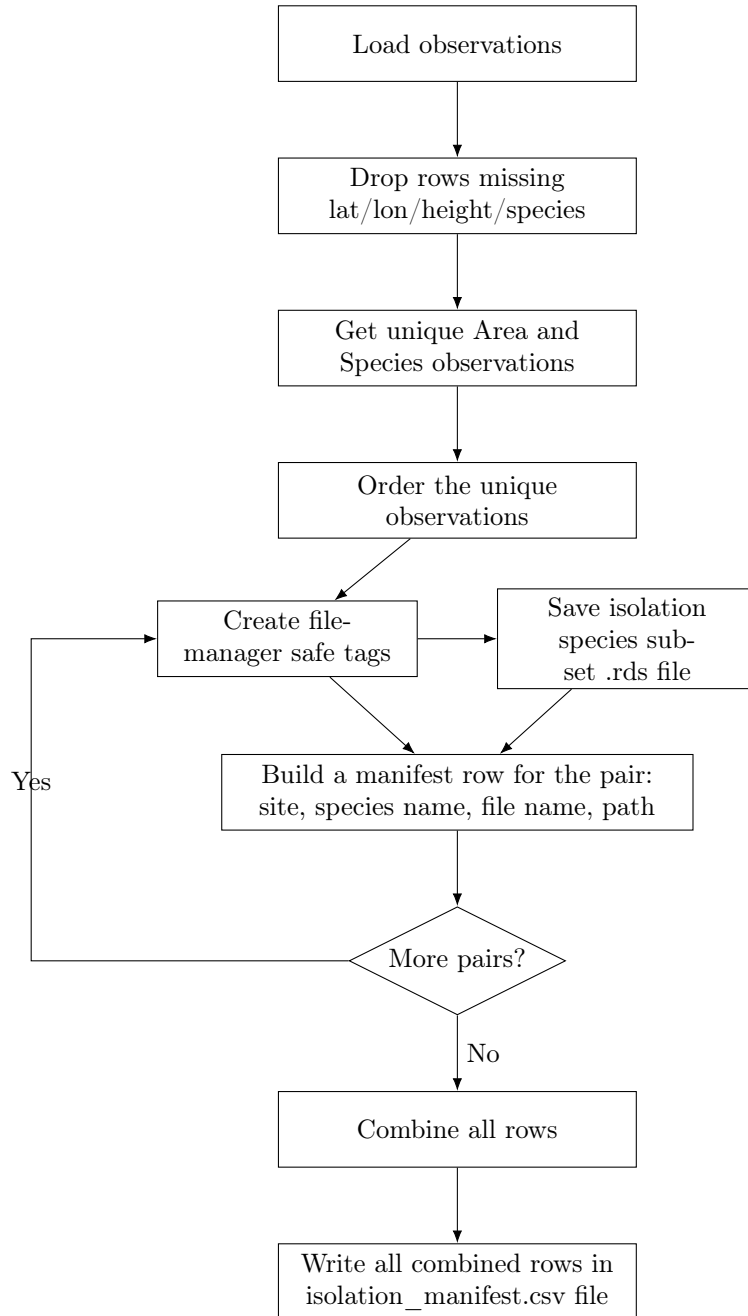


Figure 1: ...